

Reliability and Technical Acceptability of Rare-Event Detection Methods in Critical Industrial Systems

Jean-Jonathan KOFFI

Master 2 SIGLIS - U.E Research Initiation
Université de Pau et des Pays de l'Adour (UPPA)
jean-jonathan.koffi@univ-pau.fr

Abstract—Detecting rare events remains a major challenge in critical industrial systems such as aeronautics, energy, and healthcare, where even infrequent failures may lead to severe consequences. Existing approaches mainly rely on statistical methods or machine learning algorithms, whose reliability is difficult to assess due to the scarcity of representative data. This work investigates how the reliability and technical acceptability of rare-event detection methods can be ensured under extreme data limitations. Based on a structured review of recent literature on extreme value theory, rare-event simulation, imbalanced learning, and emerging AI assurance frameworks, several persistent limitations are identified, including insufficient uncertainty quantification, poorly defined cost-risk trade-offs, and the lack of robust validation processes. The study highlights the need for an integrated methodological framework combining uncertainty modeling, simulation-based validation, and explainability to improve the operational credibility of these methods in critical industrial environments. To address these issues, this work proposes an integrated methodological framework combining uncertainty modeling, simulation-based validation, and interpretability mechanisms. Rather than introducing a new detection algorithm, it provides an evaluation-oriented perspective on how rare-event detection methods can be assessed, validated, and trusted in safety-critical industrial environments.

Index Terms—Rare events, Extreme Value Theory, Imbalanced learning, Simulation, AI assurance, Critical systems

I. INTRODUCTION

Rare-event detection is an important problem in critical industrial systems such as aeronautics, energy production, transportation, and healthcare [1]. Even if abnormal situations are infrequent, they can have major consequences, from safety issues to high financial losses or environmental impact [2]. For this reason, detecting early signs of failure is not only a technical objective, but also a reliability and compliance requirement in many industrial contexts.

The core issue is simple: rare events almost never appear in the data. Most industrial datasets mainly describe normal operation, so the imbalance is structural. This means that data-driven models are trained and tested with very few examples of extreme situations. The result is often misleading: standard metrics can look reassuring, while the model still behaves badly in rare but realistic failure cases [1]. In safety-critical settings, this is exactly what makes users and engineers cautious about relying on automated decision-support.

For industrial stakeholders, accuracy alone is not the main question. What they want to know is whether the method stays reliable when uncertainty is high, whether it breaks in extreme regimes, and whether its outputs can be justified in operational terms. This is why evaluation becomes a problem on its own: the goal is not only to detect rare events, but to understand when the detection can be trusted with limited data.

Different research directions try to handle these constraints. EVT is commonly used to describe tail behavior and to estimate extreme risks when we need to go beyond what has been observed [2]. Machine learning is also widely used to detect anomalies and rare patterns in complex industrial signals, with many techniques designed for imbalanced data. Still, even when these models look strong on available datasets, their behavior in extreme or unseen conditions often remains uncertain.

To compensate for the lack of failure examples, simulation and data enrichment are often introduced to explore low-probability scenarios and to support validation [1]. At the same time, trustworthy AI and assurance frameworks add additional expectations, such as uncertainty reporting, robustness, and explainability, which become even more important in regulated industrial environments.

However, these contributions are often studied separately. In practice, statistical modeling, learning-based detection, simulation, and assurance do not naturally form a unified evaluation strategy. This makes it difficult to obtain a coherent and reliable assessment of rare-event detection methods in critical systems.

The goal of this paper is to propose a structured evaluation framework for rare-event detection under extreme data scarcity. Its main contribution is to formalize how uncertainty modeling, simulation-based validation, and explainability mechanisms can be combined to assess the reliability and technical acceptability of existing detection methods. Rather than introducing a new algorithm, this work adopts a methodological and system-level perspective focused on evaluation credibility and operational trust.

The scope of this study is deliberately generic and domain-independent. The proposed framework is intended to be adaptable to different industrial contexts and detection techniques, while remaining aligned with safety and regulatory constraints.

The remainder of this paper is organized as follows. Section II reviews the main limitations of existing approaches for rare-event detection. Section III introduces the proposed methodological architecture. Section IV presents the evaluation workflow and principles. Finally, Section V discusses industrial implications and limitations before concluding the paper.

II. STATE OF THE ART: LIMITS OF EXISTING APPROACHES

A. Rare-Event Detection in Critical Industrial Systems

Rare-event detection is a major topic in the monitoring of critical industrial systems. The difficulty comes from the fact that abnormal events are very rare, but when they happen, the consequences can be serious and sometimes irreversible. In most studies, rare events are linked to observations located in the extreme tails of probability distributions. This also explains why they are poorly represented in real datasets [3]. Even with a low frequency, these events often correspond to failure modes with high safety, economic, or environmental impact.

This problem becomes even more sensitive in domains such as aeronautics, energy production, transportation, and healthcare. In these contexts, the acceptable level of risk is extremely low, and missing a failure can lead to catastrophic outcomes. At the same time, industrial constraints make exhaustive testing unrealistic. Failure rates can be extremely small, and running real experiments is often too expensive, too slow, or simply impossible in practice [4].

For these reasons, rare-event detection should not be treated as a standard classification task. From an industrial point of view, it is closer to a reliability and risk-management problem. Classical learning and validation strategies usually assume representative datasets and frequent observations, which is not the case here. This motivates the use of dedicated modeling and evaluation approaches, discussed in the next sections

B. Statistical and Learning-Based Approaches

Statistical and machine-learning methods are widely used for anomaly and failure detection in industrial systems. They perform well for identifying deviations from normal behavior in large datasets. However, when target events are extremely rare, their reliability becomes difficult to evaluate.

A first limitation comes from the use of average performance metrics. Indicators such as accuracy or mean squared error mainly reflect behavior under normal conditions, where most data is available. As a result, rare but severe errors may remain invisible during validation. Michelucci and Venturini [1] show that a model can appear reliable while still failing in critical situations.

A second limitation is related to strong class imbalance. Rare events represent only a very small fraction of industrial datasets. Learning algorithms therefore focus on nominal patterns and tend to overlook abnormal behaviors. This phenomenon, known as the *curse of rarity* [3], leads to over-optimistic test results.

A third limitation concerns validation under data scarcity. Standard cross-validation assumes that validation data reflects

future operating conditions, which is rarely true for rare events. Moreover, limited data makes uncertainty and tail behavior difficult to estimate. As a result, operational risk is often underestimated, and failure modes may remain insufficiently characterized.

These limitations show that, although statistical and learning-based approaches are effective for detection, their evaluation remains fragile in rare-event contexts and requires complementary validation mechanisms.

C. Simulation and Data Enrichment under Scarcity

When failure examples are missing, simulation and data enrichment are often used to support learning and validation in rare-event settings. Simulation allows the generation of low-probability scenarios that are absent or poorly represented in historical datasets, making it useful for stress-testing safety-critical systems.

A common baseline is Monte Carlo sampling, where the probability of an event A is estimated as

$$\hat{P}(A) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}_A(X_i). \quad (1)$$

While this estimator is unbiased, it becomes impractical when event probabilities are extremely small, since an unrealistic number of simulations is required to observe enough rare events.

More advanced techniques, such as importance sampling and learning-assisted simulation, aim to focus computational effort on critical regions of the input space [4], [5]. However, their efficiency strongly depends on modeling assumptions and prior knowledge.

In parallel, data enrichment techniques attempt to reduce class imbalance during training. Over-sampling, generative models, and data augmentation methods can improve empirical performance [6]. Nevertheless, synthetic data does not always reflect real industrial conditions, and its representativeness remains difficult to guarantee.

The main limitation of simulation and enrichment approaches lies in their dependence on underlying models and assumptions. Without explicit validation and assurance mechanisms, their results may provide misleading confidence and remain insufficient to support technical acceptance in safety-critical systems.

D. Reliability, Uncertainty, and Open Challenges

In safety-critical industrial systems, detection performance alone is not sufficient to justify operational deployment. Industrial stakeholders must also be confident in system behavior under uncertainty, especially when safety, certification, or regulatory decisions are involved.

Rare-event detection models are often trained on limited and unrepresentative data, which makes their predictions fragile in extreme situations. When uncertainty is not explicitly quantified, a system may appear reliable while still producing high-risk decisions in rare scenarios. This issue directly affects

the balance between false alarms and missed detections, both of which can have serious operational consequences.

Regulated environments impose additional constraints on validation and certification. Goodloe [7] highlights that learning-enabled systems introduce forms of uncertainty that are not adequately addressed by traditional verification methods. Similarly, regulatory initiatives, including those of the FAA, emphasize the need for robustness analysis, interpretability, and traceable decision processes [8].

Despite these efforts, existing assurance and evaluation practices remain weakly connected to detection and simulation methods. Uncertainty modeling, validation under rare conditions, and regulatory requirements are usually addressed separately. This lack of integration limits the practical acceptability of rare-event detection systems and contributes to persistent gaps between technical performance and operational trust.

E. Synthesis of Main Limitations

Table I summarizes the main strengths and limitations of the main methodological families used for rare-event detection under extreme data scarcity.

Although each approach provides useful tools, none of them is sufficient when used alone in safety-critical industrial contexts. Extreme value theory focuses on tail modeling but relies on strong assumptions that become fragile with very limited data. Machine-learning methods are effective for complex pattern recognition, but their evaluation is often biased toward normal conditions and does not reflect extreme risks. Simulation-based approaches help explore rare scenarios, but their results strongly depend on modeling choices and computational resources. Finally, assurance frameworks support trust and certification, but they remain difficult to integrate into operational workflows.

This synthesis shows that existing approaches address complementary aspects of the problem, but leave important gaps in terms of reliability, validation, and technical acceptability. These gaps motivate the need for an integrated evaluation framework that connects modeling, learning, simulation, and assurance in a coherent way.

Approach	Main Strength	Main Limitation
Extreme Value Theory	Tail risk modeling	Strong assumptions and unstable estimates under scarcity
Machine Learning	Pattern detection in complex signals	Average bias and weak validation in extreme regimes
Simulation	Exploration of rare and critical scenarios	Strong dependency on modeling assumptions and high computational cost
Assurance Frameworks	Support for trust and certification	Limited operational integration and often conceptual nature

TABLE I
MAIN STRENGTHS AND LIMITATIONS OF EXISTING APPROACHES

F. Overview of the proposed methodological architecture

The state of the art makes one point clear: most approaches cover only part of the problem, and they are not sufficient

when taken alone. EVT is useful to model tail risk, but it relies on strong assumptions and very few extreme samples. Machine-learning models are affected by imbalance and can look convincing on standard validation while still failing in rare situations. Simulation helps explore extreme scenarios, but it often lacks a clear assurance process. As a result, reliable evaluation and technical acceptability are still difficult to guarantee in critical systems.

To address this gap, this work adopts a system-level methodological perspective instead of proposing a new detection algorithm. The objective is to give a clear structure for how rare-event detection methods can be modeled, evaluated, and justified when data is scarce. The proposed architecture is not meant to replace existing techniques. Its role is to organize them and to make their evaluation more consistent.

The architecture is built around four layers that interact with each other. The first layer focuses on rare-event modeling and tail risk, using statistical tools such as EVT. The second layer covers learning-based detection under strong imbalance, while keeping in mind that performance in extreme regimes remains uncertain. The third layer is dedicated to simulation-based validation, which explores rare and critical scenarios that are missing or under-represented in real datasets, and supports stress-testing. The last layer focuses on assurance and interpretability, including uncertainty reporting, error trade-offs, and alignment with industrial and regulatory requirements.

These layers are not independent. The goal is to connect detection performance with uncertainty awareness, validation under rare conditions, and acceptability constraints. This provides a structured way to assess rare-event detection methods beyond standard average metrics.

The next sections describe each layer and explain how they connect, before introducing the evaluation methodology based on this architecture.

Figure 1 illustrates the proposed methodological architecture and the interaction between its main components.

G. Core components of the architecture

1) *Rare-event modeling and uncertainty representation:* A central element of the proposed architecture is to explicitly model rare and extreme events, together with the uncertainty that comes with them. In critical industrial systems, the most severe risks are usually driven by extreme situations, not by average operating conditions. This is why the modeling effort must focus on the tails of probability distributions rather than on their central behavior.

Extreme Value Theory (EVT) is a statistical framework designed for this purpose, since it describes extreme events through tail modeling. Under general conditions, the distribution of normalized maxima converges to a Generalized Extreme Value (GEV) distribution. A common alternative is the peaks-over-threshold approach, where exceedances above a high threshold u are modeled using the Generalized Pareto Distribution (GPD). In that case, the conditional distribution of

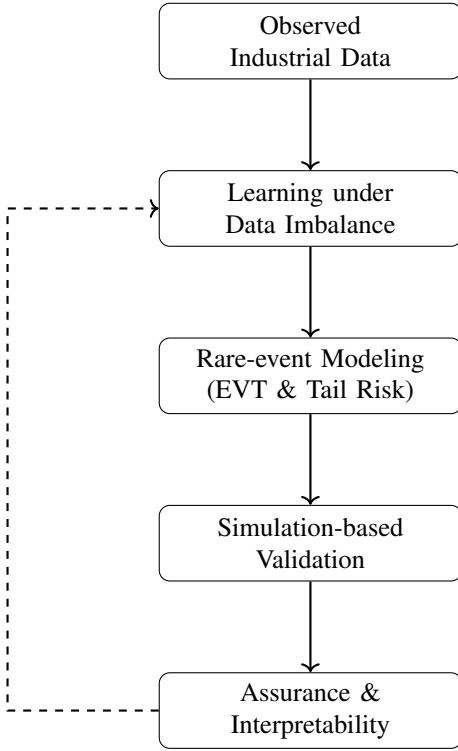


Fig. 1. Conceptual methodological architecture for rare-event detection under data scarcity

an exceedance $Y = X - u$ given $X > u$ can be approximated by

$$\mathbb{P}(Y \leq y \mid X > u) \approx 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi}, \quad (2)$$

where ξ is the tail index and β is a scale parameter. This formulation is useful to extrapolate beyond observed data and estimate extreme risk levels.

Recent work suggests that EVT can also complement learning-based detection. In particular, combining EVT with machine learning can help identify weak signals that lie in the tails of complex distributions [2]. EVT can also be used to study extreme prediction errors and rare failure regions that may be missed by models optimized for average performance [1].

At the same time, EVT has clear limitations in industrial settings. Classical EVT relies on asymptotic assumptions that may not hold when only a small number of extreme observations is available. Threshold selection and tail parameter estimation strongly influence the results and may introduce large uncertainty. Bai and Lam [9] show that EVT-based uncertainty estimates can become unstable in finite-sample regimes, which is a typical situation under severe data scarcity.

For these reasons, EVT should be viewed as an uncertainty-aware modeling tool rather than a standalone solution. In the proposed architecture, EVT-based outputs support the identification and quantification of tail risks, and they provide inputs to downstream components such as simulation-based validation and assurance layers. This interaction is needed to

assess robustness, interpretability, and technical acceptability in critical industrial systems.

2) *Learning under data scarcity and imbalance*: Machine-learning methods are widely used in modern rare-event detection because they can handle complex and high-dimensional industrial data. However, when events are rare, these methods face two strong constraints: limited data about failures and severe class imbalance.

In critical industrial systems, rare events are naturally under-represented. Most observations describe normal operation, while failure events appear only occasionally. As a result, standard learning algorithms tend to focus on the majority class and optimize average metrics, which often leads to low sensitivity to rare but safety-critical events [3]. This is commonly described as the *curse of rarity*: models can look accurate during validation while still failing in extreme conditions once deployed.

To reduce imbalance effects, several strategies exist, including cost-sensitive learning, ensemble methods, and adapted loss functions. Optimization-based ensemble weighting methods are a good example, since they explicitly account for imbalance during training [10]. Still, these approaches often depend on heuristic cost choices and they do not solve the core issue: the lack of information in extreme regions of the data space.

Model evaluation is another limitation. Metrics such as accuracy, precision, recall, or F1-score provide only a partial view of behavior and usually do not capture the risk associated with extreme errors. Under severe imbalance, validation results may therefore look reassuring while hiding important failure modes [1].

From an architectural perspective, machine learning should be treated as a pattern-recognition component rather than a standalone solution. In the proposed framework, learning-based models interact with extreme value modeling to better capture tail behaviors, and with simulation-based validation to assess robustness beyond observed data. This makes it possible to use machine learning while explicitly accounting for imbalance, uncertainty, and rare-event risk.

3) *Simulation-based validation layer*: Since rare and extreme events occur very infrequently, real-world datasets are often not enough to evaluate detection methods under truly critical conditions. This is why simulation-based approaches are important in the proposed architecture: they allow the exploration of low-probability scenarios that are missing or poorly represented in observed data.

Monte Carlo simulation is a standard baseline for system evaluation using repeated random sampling. However, for rare-event analysis, naive Monte Carlo quickly becomes inefficient because an extremely large number of simulations is required to observe enough critical events. This motivates rare-event simulation methods that focus computational effort on low-probability but high-impact regions of the input space.

Importance sampling is one of the most common techniques in this context. It modifies the sampling distribution to increase the occurrence of rare events, while keeping unbiased risk

estimation through sample reweighting. It has been applied successfully in safety-critical domains, for example to optimize maintenance intervals in redundant subsystems with very low failure rates [4].

More recent work combines simulation with learning-based models. Deep Probabilistic Accelerated Evaluation (Deep-PrAE), for instance, integrates rare-event simulation principles with neural networks to identify critical regions in complex black-box systems [5]. This shows that simulation can support both evaluation and targeted exploration of failure-inducing scenarios in high-dimensional settings.

Generic simulation frameworks also exist for reliability and availability analysis in complex industrial systems. AvailSim4 combines Monte Carlo and discrete-event simulation to model interdependent failures and estimate system-level risk indicators [11]. Even if such tools are not designed specifically for learning-based detection, they remain useful for risk-oriented evaluation under industrial constraints.

Simulation-based validation also has limits. Results depend strongly on modeling assumptions and system fidelity, and simulated behaviors do not always transfer to real operational environments. In addition, simulation alone does not guarantee technical acceptability or regulatory compliance, since it does not directly address uncertainty, interpretability, or residual risk.

For these reasons, simulation-based validation is treated as a complementary layer in the proposed architecture. It supports stress-testing and rare-event exploration, while remaining connected to uncertainty modeling and assurance mechanisms. This integration is needed so that simulation evidence contributes meaningfully to reliability assessment and industrial acceptability.

4) *Assurance and interpretability layer*: In safety-critical industrial systems, rare-event detection cannot be justified by predictive performance alone. Even if accuracy looks good, industrial stakeholders and regulators still need confidence in how the system behaves under uncertainty and extreme operating conditions. This is why assurance and interpretability are treated as a core layer in the proposed architecture.

A first requirement for technical acceptability is to make uncertainty explicit and understandable. Detection outputs should not only report a decision, but also provide information about confidence, uncertainty bounds, and potential failure modes. This point becomes essential in rare-event contexts, where predictions often involve extrapolation beyond observed data. Without uncertainty awareness, automated outputs may look reliable while being misleading in critical situations.

Another major issue is the error trade-off. In rare-event detection, false negatives can have severe consequences, while false positives can trigger unnecessary actions, disrupt operations, and increase costs. The acceptable balance between these errors depends on system criticality and operational constraints, and it cannot be reduced to a standard learning objective. For this reason, these trade-offs must be addressed at the system level.

Recent work on safety-critical machine-learning systems highlights the need for structured assurance processes. Goodloe [7] shows that learning-enabled components introduce uncertainties that are not fully covered by traditional software verification. Institutional initiatives, including the Federal Aviation Administration's AI safety assurance roadmap, also emphasize progressive deployment, transparency, and consistency with certification practices [8].

Broader assurance frameworks support the same direction. Claims-based approaches propose to define explicit safety and reliability claims and to support them with evidence from modeling, simulation, testing, and operational monitoring [12]. In this view, technical acceptability is not defined by a single metric, but by the consistency between system claims, supporting evidence, and residual risk management.

Interpretability is also essential. Being able to explain model behavior, identify influential factors, and justify decisions is important for human oversight, debugging, and regulatory review. In rare-event detection, interpretability can also help highlight weak signals and possible failure precursors, which supports trust and better decision-making.

In the proposed architecture, the assurance and interpretability layer acts as a link between detection, modeling, and validation. It ensures that uncertainty, risk trade-offs, and regulatory constraints are explicitly considered, which is necessary to support technical acceptability in critical industrial systems.

H. Interaction between components

The proposed architecture is not effective because of one single component taken alone, but because the components work together. In critical industrial systems, rare-event detection requires coordination between modeling, learning, simulation, and assurance layers to keep the overall evaluation consistent and credible.

Learning-based components work on observed industrial data to detect patterns, anomalies, or weak signals. In rare-event settings, the main issue is that the data is heavily imbalanced and contains very few failure examples. For this reason, model outputs should be treated with caution: uncertainty is part of the problem. EVT complements learning by looking specifically at tail behavior and extreme deviations. This helps identify high-risk regions that can be missed when models are mainly optimized for average performance. EVT outputs can therefore be used as uncertainty-aware indicators to support both detection confidence and validation.

Simulation-based validation plays a different role. It links modeling assumptions to operational robustness by exploring rare and critical scenarios that are unlikely to appear in real datasets. In practice, this makes stress-testing possible and allows sensitivity studies on key parameters. It is also important to keep a strict separation between observed data used for learning and synthetic data generated for validation, so that simulation improves evaluation rather than artificially improving apparent performance.

The assurance and interpretability layer then brings these elements together. It consolidates uncertainty estimates (from

EVT), learning outputs, and robustness evidence (from simulation) to support risk-oriented decision-making. The objective is to make confidence levels, error trade-offs, and residual risks explicit, and to keep the evaluation aligned with safety goals and regulatory constraints.

Finally, the architecture includes feedback loops. Evidence obtained during simulation and assurance can lead to model refinements, threshold adjustments, or updated detection strategies. For instance, when stress-testing reveals specific failure modes or high-risk regions, this can motivate changes in learning objectives or EVT settings. This iterative process supports improvement while maintaining traceability and transparency.

Overall, this organization treats rare-event detection as an integrated process rather than a purely predictive task. This system-level view is required to obtain reliable and technically acceptable solutions in safety-critical industrial environments.

I. Positioning with respect to industrial constraints

Critical industrial systems come with practical constraints that shape how rare-event detection can be designed, validated, and deployed. Safety requirements, regulatory rules, limited access to failure data, and transparency expectations all matter. In these contexts, performance alone is not enough, and the proposed architecture is built with these constraints in mind.

A first issue is simply the lack of representative data, especially for rare and catastrophic events. Unlike research benchmarks, industrial teams rarely have large labeled datasets that contain enough failures for training and validation. The proposed architecture deals with this by separating learning from validation, and by using simulation to explore extreme scenarios instead of relying only on observed failures.

Regulatory and certification processes also require strong justification. In domains such as aeronautics, energy, or healthcare, learning-based systems are expected to be introduced progressively and supported by evidence of reliability, robustness, and risk control. The proposed architecture follows this logic by emphasizing uncertainty quantification, traceability, and claims-based assurance rather than black-box optimization. This is consistent with institutional roadmaps that promote gradual deployment and continuous monitoring of AI-enabled systems.

Interpretability also matters in practice. Detection outputs must be understandable and actionable, especially when human operators remain responsible for safety-critical decisions. By treating interpretability and uncertainty reporting as core elements, the framework supports human oversight and makes operational and regulatory review easier.

Cost and operational constraints also limit large validation campaigns. Large-scale real testing is often unrealistic because of time, resources, or safety restrictions. Simulation-based validation and rare-event-focused evaluation provide a practical way to focus on high-risk scenarios without relying only on expensive experiments. Still, simulation cannot replace real operational evidence and should be used as a complement within a broader assurance process.

Finally, the proposed architecture is designed to stay generic and adaptable. It is not tied to one industrial domain or one detection algorithm. Instead, it provides a methodological structure that can be adapted to different contexts and regulatory environments, while keeping a consistent approach to reliability assessment and technical acceptability.

By explicitly incorporating industrial constraints, the proposed architecture offers a structured basis for rare-event detection in real-world systems. The next section builds on this framework with a concrete methodology for assessing and validating rare-event detection under extreme data scarcity.

III. PROPOSAL

A. Objectives and scope of the proposed methodology

This work proposes a structured methodology to assess the reliability and technical acceptability of rare-event detection methods in critical industrial systems with extreme data scarcity. The contribution is not a new detection algorithm, but a way to consistently and risk-consciously combine and evaluate existing statistical tools, learning-based methods, and simulation techniques. The first goal is to go beyond standard performance-driven validation. In rare-event settings, classical evaluation strategies can be misleading because failure data is limited or even unavailable. The proposed methodology therefore aims to support evaluation practices that explicitly account for uncertainty, extreme risks, and real industrial constraints.

A second goal is to make detection systems acceptable from an industrial and regulatory perspective. This cannot be achieved with good scores alone. Evaluation also needs to provide transparency, interpretability, and traceability. In practice, stakeholders want to know not only whether a method works on the available data, but also when its outputs can be trusted and where it may fail.

The scope of the methodology is kept generic on purpose. It is intended to apply across critical sectors such as aeronautics, energy, transportation, and healthcare, without depending on a specific industrial process or a specific machine-learning architecture. It also does not assume large labeled datasets or frequent failure observations, which matches many real industrial settings where rare events are infrequent by design.

The proposal is based on a few realistic assumptions. Rare-event detection is treated as decision-support, meaning that human operators remain responsible for safety-critical decisions. Simulation models are assumed to be available to approximate system behavior under extreme conditions, even if they are imperfect. Finally, uncertainty is assumed to be quantifiable and communicable in a way that helps risk-aware evaluation.

By clarifying these objectives and assumptions, the proposed approach provides a realistic foundation for the integrated framework presented in the next sections. The next subsection introduces the methodological principles that guide the framework.

B. Methodological principles

The proposed methodology is based on a small set of evaluation principles that come directly from the limitations discussed in the state of the art and from real industrial constraints. The goal is not to define how rare-event detection methods should be built, but to clarify how they should be assessed in order to obtain results that remain reliable and technically acceptable under extreme data scarcity.

A first principle is to clearly separate learning from validation. In rare-event settings, using the same limited data for both training and evaluation can produce over-optimistic results. The proposed approach therefore separates the data used to train detection models from the mechanisms used to assess them, especially for rare and extreme situations. This prevents confusing good performance on nominal data with true robustness under low-probability conditions.

A second principle is to focus on extreme risk rather than average performance. Conventional evaluation often uses mean-based indicators, which reveal little about rare but high-impact failures. Here, the emphasis is on extreme errors, tail behaviors, and critical scenarios, since these matter most in safety-critical systems. Uncertainty awareness is also treated as a core element of evaluation. Under severe scarcity, both predictions and validation outcomes remain uncertain. For this reason, uncertainty indicators are considered part of the evaluation itself, so that results can be interpreted in a risk-aware way instead of being treated as definitive conclusions.

Another important point is to keep a clear distinction between observed and simulated data. Simulation is necessary to explore rare and extreme scenarios, but stress-testing evidence should not be treated as equivalent to real operational observations. Maintaining this separation helps avoid artificial confidence in detection performance.

The methodology follows a progressive, iterative validation logic. Since exhaustive testing is rarely feasible in industrial settings, validation relies on incremental evidence. Results are used to refine assumptions and reassess risk, while maintaining traceability and transparency.

Together, these principles provide a consistent basis for the integrated workflow described next.

C. Integrated workflow for rare-event detection and validation

This subsection introduces an integrated evaluation workflow for rare-event detection methods in critical industrial systems. The workflow is generic and algorithm-agnostic, and it is designed to support reliability assessment and technical acceptability under extreme data scarcity.

The workflow starts from available operational data, with the acknowledgement that rare or catastrophic events may be limited or even absent. Data preparation aims to preserve information related to extreme behaviors while keeping realistic distributions. Artificial balancing or frequency distortion is avoided in order to remain consistent with real operating conditions.

Learning-based models are first applied to observed data to characterize normal behavior and flag possible anomalies. In

rare-event settings, datasets are imbalanced and contain few extreme examples, so model outputs should not be treated as final decisions. The aim at this stage is to surface signals for further investigation, not to optimize a single predictive score.

An extreme-risk analysis is then performed on selected variables or model outputs, focusing on tail behavior and extreme deviations missed by average-based metrics. This step highlights potentially high-risk situations and provides additional input for validation.

Simulation-based validation is used to probe further. Because rare, critical scenarios are unlikely to appear in real data, simulation enables exploration of stress conditions and robustness under extreme regimes. Simulated scenarios serve as validation evidence, but are not treated as equivalent to real operational observations.

Evaluation results are then integrated from an assurance perspective. Learning outcomes, extreme-risk analysis, and simulation evidence are considered together to discuss uncertainty, error trade-offs, and residual risk. The goal is an interpretable assessment relevant for industrial decision-making, rather than a single aggregate metric.

Finally, the workflow supports refinement. Insights from stress-testing and assurance review can lead to updated modeling assumptions, detection thresholds, or evaluation criteria. This iterative process enables gradual improvement while keeping evaluation aligned with industrial constraints.

By structuring evaluation as an integrated and uncertainty-aware workflow, the proposed methodology provides a coherent basis to assess rare-event detection methods in safety-critical environments, even when data availability is severely limited.

D. Uncertainty-aware and risk-oriented evaluation criteria

Evaluating rare-event detection methods in critical industrial systems requires more than standard performance metrics. When events are extremely infrequent but potentially catastrophic, average-based indicators such as accuracy, mean squared error, or global F1-score provide limited insight into reliability and can hide important risks.

The proposed methodology evaluates models through uncertainty and risk. Rather than relying on point estimates, it emphasizes uncertainty in predictions and validation, using confidence intervals, sensitivity analyses, and uncertainty bounds to assess robustness, especially when extrapolating beyond observed data.

Risk-oriented evaluation also examines extreme errors and tail behavior via extreme value modeling, focusing on the most severe prediction errors or deviations. This highlights worst-case or near-worst-case scenarios, which are more relevant for safety-critical applications than metrics reflecting only average behavior.

Another criterion is the treatment of false positives and false negatives. In rare-event detection, these errors are asymmetric: missing a critical event can be catastrophic, whereas excessive false alarms cause unnecessary actions, operational disruption, and loss of trust. The framework treats this trade-off as

context-dependent, allowing acceptable error balances to be set by system criticality and operational constraints.

Scenario-based evaluation further analyzes model behavior in simulated rare and extreme conditions, checking detection performance under stress and the consistency of uncertainty estimates. This reveals failure modes, boundary conditions, and limitations that aggregate metrics may hide.

Together, these criteria provide a realistic assessment of rare-event detection methods. By combining uncertainty quantification, extreme-risk analysis, and context-aware error trade-offs, the methodology supports better decisions and technical acceptability in safety-critical environments.

E. Technical acceptability and assurance perspective

In critical industrial environments, rare-event detection methods cannot be deployed based on empirical performance alone. Technical acceptability depends on showing that the system behaves in a controlled way under uncertainty, that its limitations are known, and that residual risks are explicitly identified and managed. The proposed methodology addresses these requirements through an assurance-oriented perspective.

A central idea is evidence-based evaluation. Instead of relying on a single indicator, the methodology combines complementary evidence coming from learning behavior, uncertainty analysis, and simulation-based stress testing. This makes it possible to form a more informed judgment about how the system behaves in rare and extreme situations, which is closer to industrial assurance practices.

Transparency and traceability are also required in practice. Stakeholders need to understand how conclusions are obtained, which assumptions are made, and under which conditions the system could fail. By integrating uncertainty quantification and interpretability into evaluation, the proposed approach supports decision-support that remains traceable rather than opaque.

Regulatory and institutional frameworks increasingly emphasize structured assurance processes for learning-enabled systems. Guidance from organizations such as NASA and the Federal Aviation Administration highlights progressive deployment, continuous monitoring, and consistency with certification practices. Claims-based assurance approaches also promote the idea of framing safety and reliability as explicit claims supported by evidence throughout the system lifecycle. The proposed methodology follows this direction by focusing on progressive validation and explicit risk management rather than binary acceptance.

Human oversight remains essential. In this framework, rare-event detection systems are treated as decision-support tools, not autonomous decision-makers. Uncertainty-aware outputs and interpretable indicators help operators assess risk, apply judgment, and intervene when needed, which is necessary in high-stakes environments.

Finally, the methodology makes residual risk explicit. By exposing uncertainty, assumptions, and potential failure modes, the proposed approach supports realistic expectations about system capabilities. This transparency is important to

maintain trust and to deploy rare-event detection methods responsibly under industrial and regulatory constraints.

F. Assumptions, limitations, and expected benefits

The proposed framework relies on a few explicit assumptions. First, it assumes that rare-event detection systems operate in contexts where data describing extreme or catastrophic failures is scarce or even unavailable, which is typical in safety-critical industrial systems. Second, it assumes that simulation models can be built to approximate system behavior under extreme conditions, while recognizing that simulations remain imperfect representations of reality. Finally, it assumes that uncertainty can be quantified and communicated in a way that supports human judgment and decision-making.

Even if the approach is designed to remain generic, it still has limitations. Simulation-based validation depends on the quality of the simulation models and on the assumptions behind them. In addition, simulated scenarios do not always capture the full complexity of real operations. Uncertainty estimates and extreme-risk indicators can also change significantly depending on modeling choices, parameter settings, and simplifications. For this reason, the methodology does not claim to eliminate risk. Its purpose is to make residual risk visible and easier to track and manage.

Another limitation is the absence of formal performance guarantees. The contribution focuses on methodological structure and evaluation practices, not on theoretical bounds or optimality results. This choice fits industrial needs, but it also means that results should be interpreted as contextual evidence rather than as absolute guarantees.

Despite these limits, the methodology brings practical value. By combining learning-based detection, extreme-event modeling, simulation-based validation, and assurance considerations, it provides a structured way to evaluate rare-event detection methods under extreme data scarcity. It supports more realistic risk assessment, improves transparency, and helps align evaluation with industrial and regulatory expectations. Overall, this can strengthen trust and support responsible deployment in critical industrial environments.

G. Discussion and methodological implications

This work does not aim to show better predictive performance than existing methods. Instead, it clarifies how rare-event detection approaches should be evaluated when data is scarce and risks are high. The main contribution is therefore methodological: it focuses on how reliability and technical acceptability can be assessed, rather than on designing a new detection algorithm.

One important point concerns extreme risk. Many evaluation practices still rely on average-based indicators, which can hide critical failure behaviors. By focusing evaluation on extreme deviations, uncertainty, and stress scenarios, the proposed approach moves attention away from overall accuracy and toward risk exposure. This leads to a more realistic interpretation of performance in safety-critical environments, where rare failures matter more than frequent correct predictions.

Another point is that isolated validation strategies are not enough. Learning-based detection, statistical modeling, and simulation are often used separately, each with its own assumptions and limitations. Under rare-event conditions, none of these tools is sufficient on its own to support a credible evaluation. Their interest comes from how they are combined: learning outputs are treated as intermediate signals, uncertainty is made explicit, and simulation is used to challenge model behavior rather than to make results look better.

The methodology also reduces the risk of over-optimistic conclusions. In rare-event contexts, validation can appear convincing simply because extreme situations are missing from the dataset. By separating learning from validation and by keeping observed data distinct from simulated stress-testing evidence, the approach encourages more conservative and transparent evaluation. This helps avoid deploying systems that perform well on paper but fail in rare and critical conditions.

From an industrial point of view, the framework also helps connect technical evaluation with operational acceptability. Outputs are not limited to performance scores: they include uncertainty information, risk trade-offs, and explicit assumptions. This better fits industrial decision-making, where stakeholders need to understand not only how a system performs, but also when its outputs can be trusted.

Overall, the contribution remains methodological rather than algorithmic. The framework does not replace existing rare-event detection techniques. It provides a consistent way to assess them under data scarcity and high risk. By treating evaluation as an integrated process instead of a final metric, the methodology supports a more credible and responsible use of rare-event detection systems in critical industrial settings.

H. Positioning with respect to existing approaches

The proposed methodology is not meant to replace existing rare-event detection approaches. It is intended as a complementary evaluation framework. The contribution is mainly about how different families of methods can be assessed and combined in practice, rather than about introducing a new detection mechanism.

Statistical methods based on extreme value modeling help analyze tail behavior and rare deviations but, used alone, rely on strong assumptions and are sensitive to limited data. In the proposed framework, they are not definitive risk estimators but uncertainty-aware indicators that must be validated and interpreted with other evidence.

Learning-based detection methods flexibly model complex industrial data, yet are often judged by average-oriented metrics that underestimate rare, high-impact failures. Here, their outputs are treated as intermediate signals whose reliability is evaluated through uncertainty analysis and stress-testing, rather than isolated validation scores.

Simulation-based approaches are also widely used to explore rare scenarios when empirical data is insufficient. However, simulation is often applied as a standalone analysis tool. In the proposed methodology, simulation is explicitly positioned as a validation layer. Its role is to challenge detection

behavior and support evidence accumulation, while remaining clearly separated from observed operational data.

Assurance-oriented frameworks promote transparency, traceability, and risk management in learning-enabled systems but often stay conceptual. The proposed methodology operationalizes these principles through practical evaluation steps that combine uncertainty modeling, simulation-based validation, and risk-oriented interpretation.

This work is methodological, not algorithmic. It offers a structured way to evaluate and compare rare-event detection methods under extreme data scarcity, without favoring any modeling paradigm. By emphasizing evaluation coherence and technical acceptability, it fills gaps not addressed by existing approaches considered individually.

I. Applicability to critical industrial systems

The proposed methodology is designed for critical industrial systems where rare events are both highly consequential and poorly represented in available data. Its applicability does not depend on domain-specific models, but on evaluation principles that are common across many safety-critical contexts. In practice, catastrophic events are rare in these systems, which makes training and validation difficult. Most of the time, there are simply not enough failure examples to build a representative dataset. The proposed framework deals with this by separating learning from validation, and by using uncertainty-aware indicators and simulation-based stress-testing instead of relying only on observed failure frequency. This is important because not observing failures does not automatically mean that the system is risk-free.

Another aspect of applicability is compatibility with industrial decision-making. In most critical systems, detection tools support human operators instead of replacing them. The methodology follows this reality by emphasizing interpretability, uncertainty communication, and risk-oriented evaluation, so that detection outputs can be integrated into operational and safety workflows.

From a regulatory and certification point of view, the methodology supports progressive validation instead of a simple accept/reject decision. The idea is to build confidence step by step, using evidence from observed data, extreme-risk analysis, and simulated scenarios. This fits industrial practices where systems are introduced gradually, with traceability and regular risk review.

The methodology is also meant to stay adaptable. It can be applied to different sectors (aeronautics, energy, transportation, healthcare) without changing its main structure. The goal is not to optimize performance for one specific application, but to provide a practical way to assess reliability and technical acceptability across safety-critical systems.

Overall, the approach remains focused on evaluation under real constraints rather than ideal conditions. This makes it relevant for industrial contexts where rare-event detection must be assessed carefully, even with limited data.

IV. CONCLUSION

This paper discussed how to evaluate rare-event detection methods in critical industrial systems, where failures can have major consequences but are poorly represented in available data. The contribution is not a new detection algorithm. Instead, it proposes a methodological framework to assess existing approaches under extreme data scarcity. By combining uncertainty-aware modeling, learning under imbalance, simulation-based stress testing, and an assurance-oriented view, the approach shifts attention away from average performance and toward reliability under risk and uncertainty.

From an industrial point of view, this perspective is especially relevant in safety-critical environments where detection systems act as decision-support tools rather than fully autonomous controllers. The framework promotes evaluation practices that remain compatible with industrial and regulatory constraints, including transparency, traceability, and progressive validation. By separating learning from validation, and observed data from simulated evidence, it helps avoid over-optimistic conclusions and supports more realistic judgments about technical acceptability.

Future work could focus on a few concrete extensions. One is to strengthen the connection between rare-event evaluation and formal assurance or certification frameworks. Another is to improve uncertainty quantification under severe data scarcity, since this remains a weak point in many practical settings. It would also be interesting to study hybrid validation strategies that combine simulation, operational data, and expert knowledge. Finally, applying the methodology to real industrial case studies would help clarify its practical benefits and limitations. Overall, this work underlines the importance of risk-oriented evaluation when rare-event detection systems are deployed in critical industrial environments.

REFERENCES

- [1] U. Michelucci and F. Venturini, “New Statistical Framework for Extreme Error Probability in High-Stakes Domains for Reliable Machine Learning,” Mar. 2025, arXiv:2503.24262 [cs]. [Online]. Available: <http://arxiv.org/abs/2503.24262>
- [2] S. Cl  men  on and A. Sabourin, “Weak Signals and Heavy Tails: Machine-learning meets Extreme Value Theory,” Jun. 2025, arXiv:2504.06984 [math]. [Online]. Available: <http://arxiv.org/abs/2504.06984>
- [3] C. Shyalika, R. Wickramarachchi, and A. Sheth, “A Comprehensive Survey on Rare Event Prediction,” Oct. 2024, arXiv:2309.11356 [cs]. [Online]. Available: <http://arxiv.org/abs/2309.11356>
- [4] R. Pfaff, K. Melcher, and J. Franzen, “Rare Event Simulation to Optimise Maintenance Intervals of Safety Critical Redundant Subsystems,” *PHM Society European Conference*, vol. 4, no. 1, Jun. 2018. [Online]. Available: <https://papers.phmsociety.org/index.php/phme/article/view/433>
- [5] M. Arief, Y. Bai, W. Ding, S. He, Z. Huang, H. Lam, and D. Zhao, “Certifiable Deep Importance Sampling for Rare-Event Simulation of Black-Box Systems,” Nov. 2021, arXiv:2111.02204 [stat]. [Online]. Available: <http://arxiv.org/abs/2111.02204>
- [6] J. Hua, R. Ahluwalia, and R. Chandra, “Extreme value forecasting using relevance-based data augmentation with deep learning models,” Oct. 2025, arXiv:2510.02407 [cs]. [Online]. Available: <http://arxiv.org/abs/2510.02407>
- [7] A. E. Goodloe, “Assuring Safety-Critical Machine Learning Enabled Systems: Challenges and Promise.”
- [8] R. L. P. Aff  t, “AI Risk Management Frameworks,” 2025.
- [9] Y. Bai and H. Lam, “Uncertainty Quantification and Confidence Intervals for Naive Rare-Event Estimators,” *J. Appl. Probab.*, vol. 62, no. 1, pp. 84–110, Mar. 2025, arXiv:2305.02434 [stat]. [Online]. Available: <http://arxiv.org/abs/2305.02434>
- [10] G. Tertytchny, G. L. Stavrinides, and M. K. Michael, “Rare Event Detection in Imbalanced Multi-Class Datasets Using an Optimal MIP-Based Ensemble Weighting Approach,” *AAAI*, vol. 39, no. 19, pp. 20 867–20 875, Apr. 2025, arXiv:2412.13439 [cs]. [Online]. Available: <http://arxiv.org/abs/2412.13439>
- [11] M. Blaszkiewicz, A. Apollonio, T. Buffet, T. Cartier-Michaud, L. Felsberger, J. Heron, J. Uythoven, and D. Wollmann, “AvailSim4: Open Source Framework For Availability And Reliability Simulations.”
- [12] A. S. Kapusta, D. Jin, P. M. Teague, R. A. Houston, J. B. Elliott, G. Y. Park, and S. S. Holdren, “A Framework for the Assurance of AI-Enabled Systems,” Apr. 2025, arXiv:2504.16937 [cs]. [Online]. Available: <http://arxiv.org/abs/2504.16937>